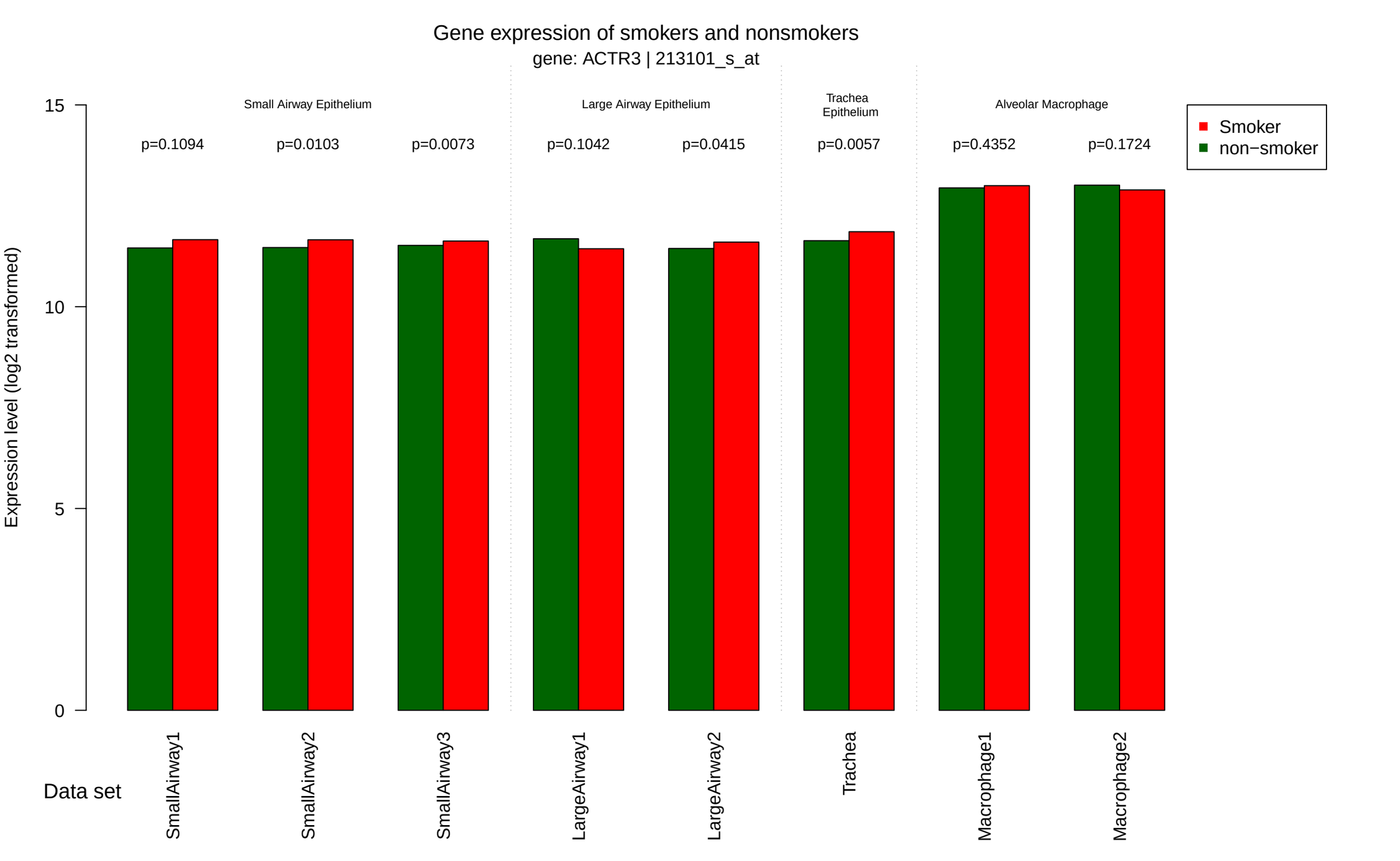
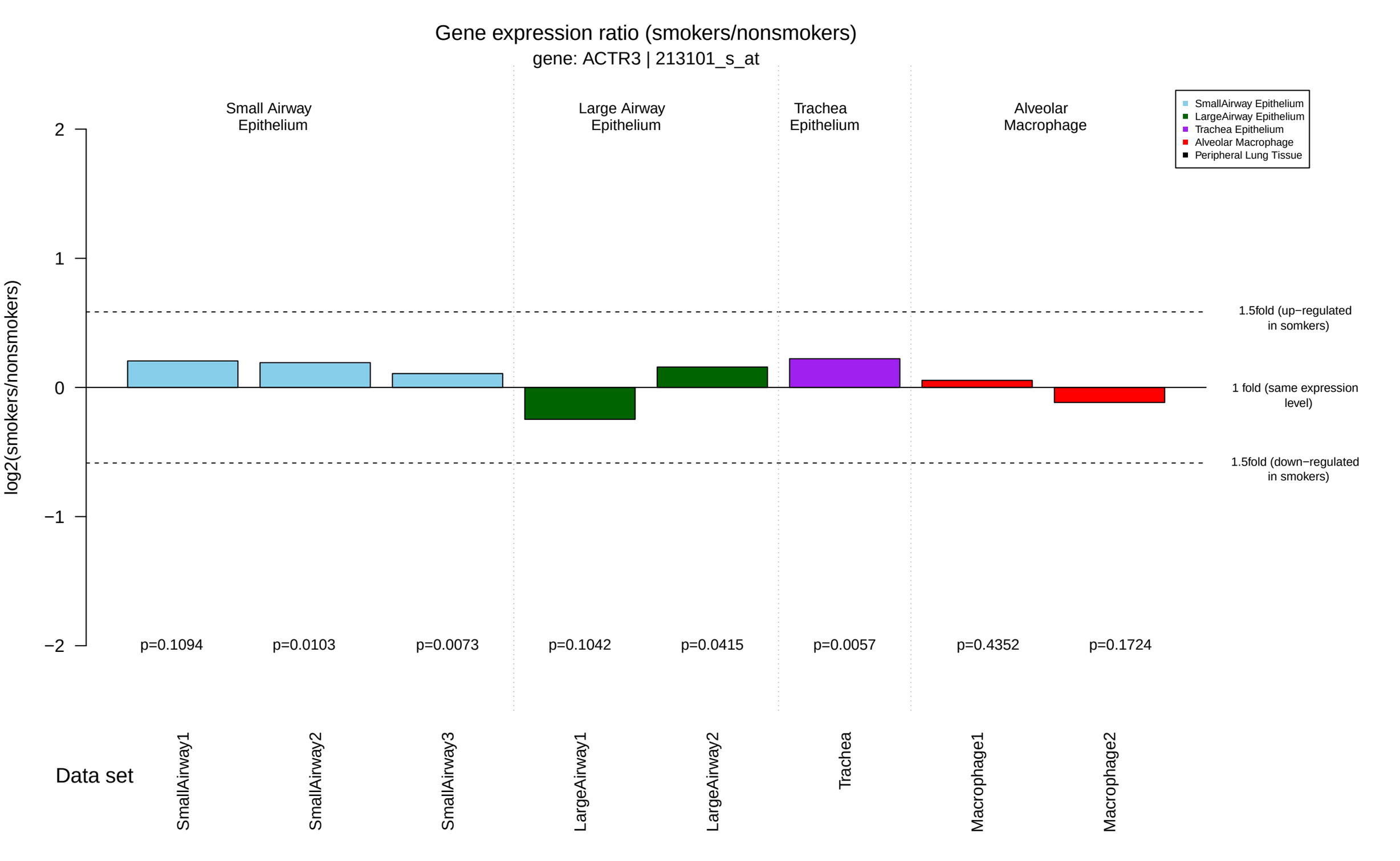
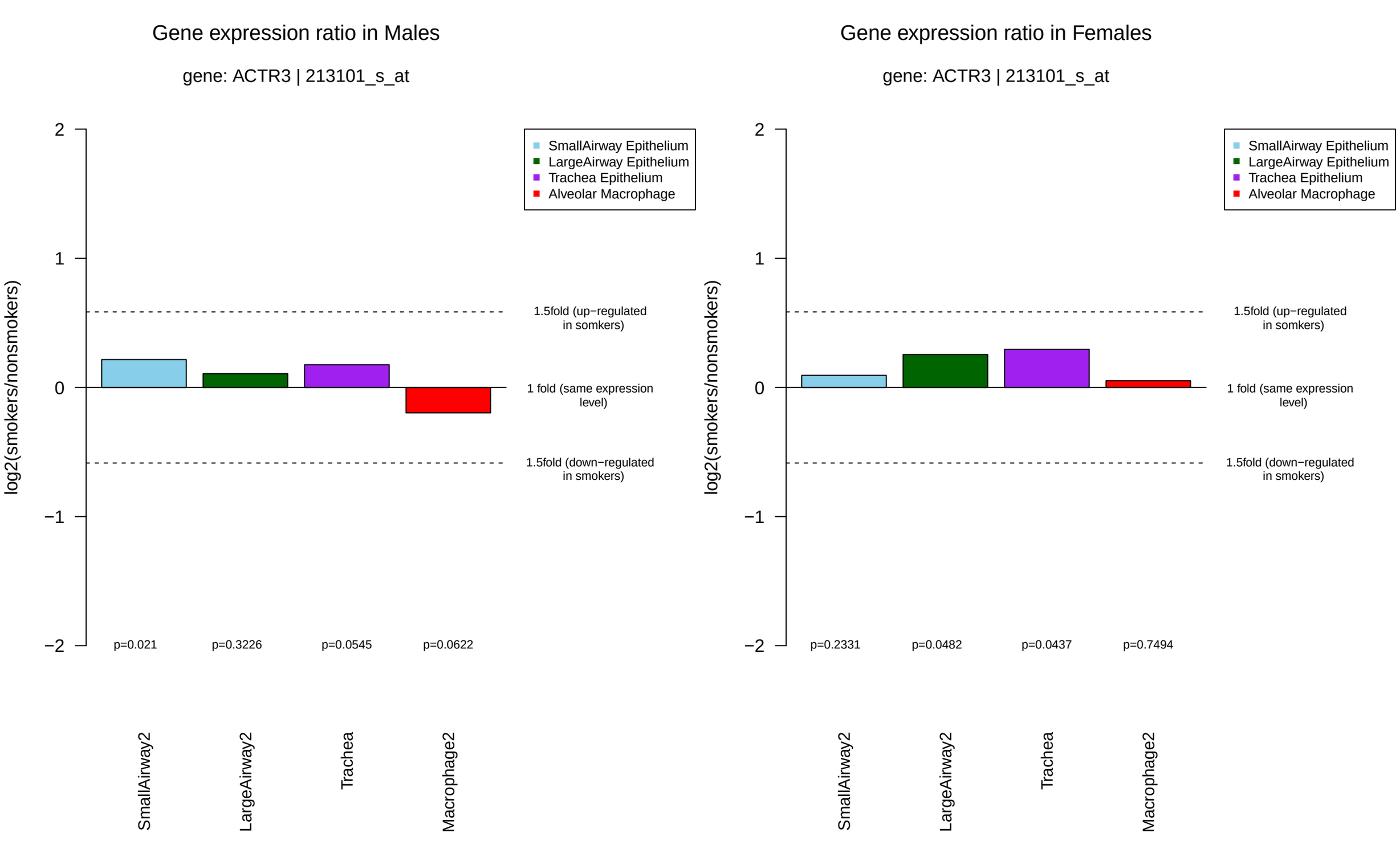
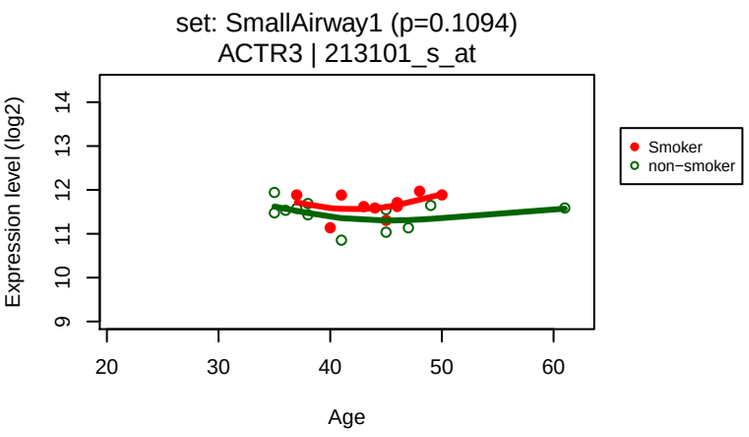
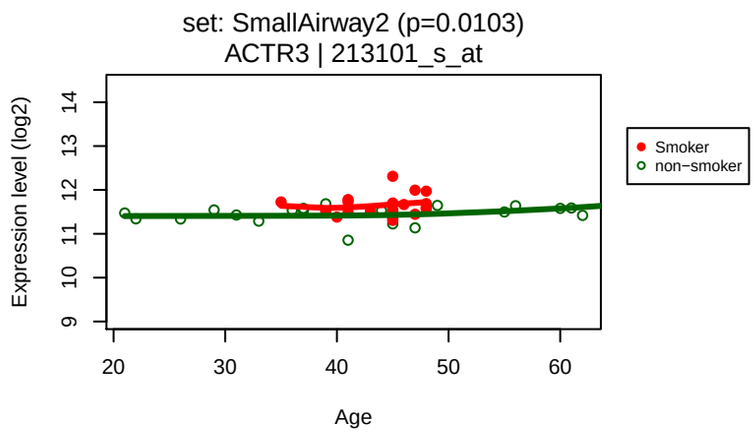
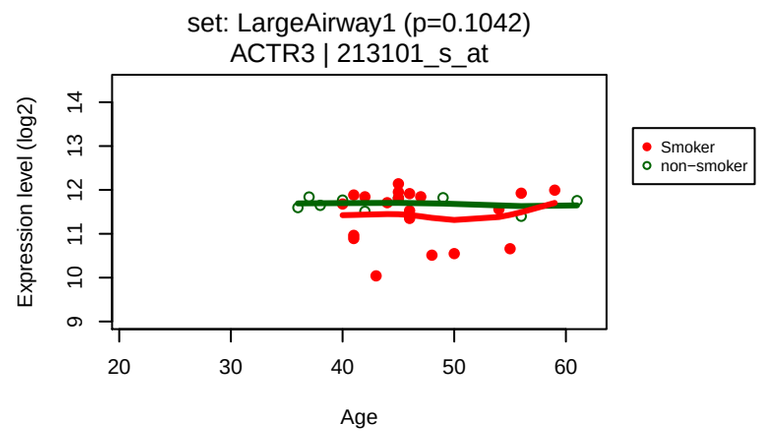

Gene expression of smokers and nonsmokers
gene: ACTR3 | 213101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1094
p=0.0103
p=0.0073
p=0.1042
p=0.0415
p=0.0057
p=0.4352
p=0.1724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

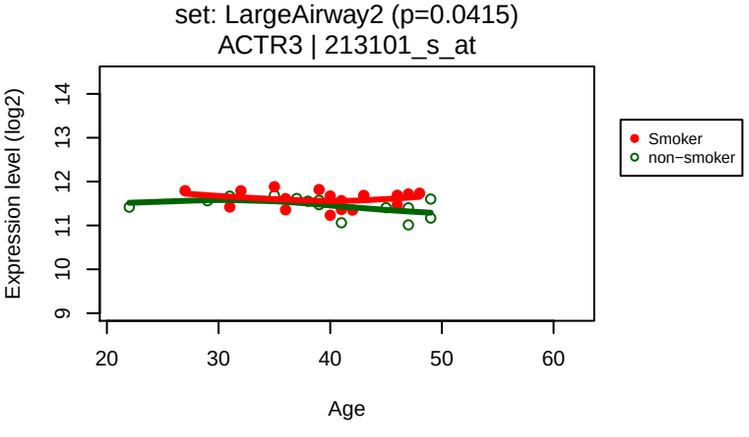

Gene expression ratio (smokers/nonsmokers)
gene: ACTR3 | 213101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1094
p=0.0103
p=0.0073
p=0.1042
p=0.0415
p=0.0057
p=0.4352
p=0.1724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

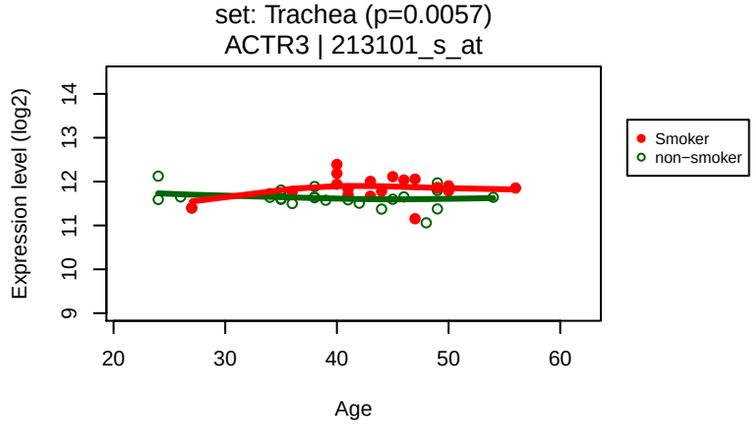

Gene expression ratio in Males
gene: ACTR3 | 213101_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.021
p=0.3226
p=0.0545
p=0.0622
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR3 | 213101_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2331
p=0.0482
p=0.0437
p=0.7494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

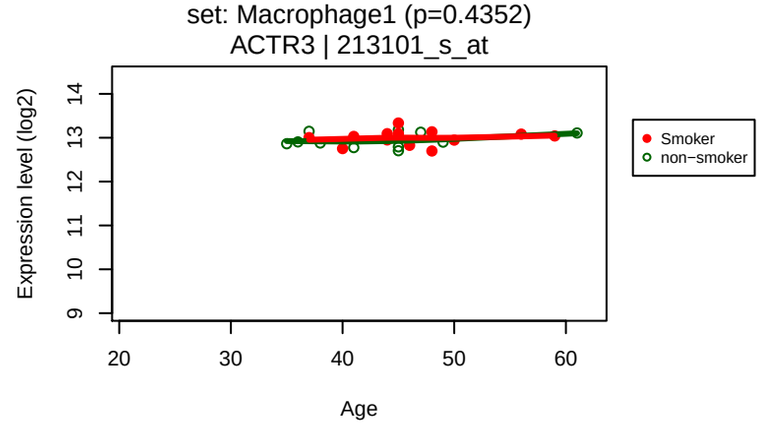

set: SmallAirway1 (p=0.1094)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker

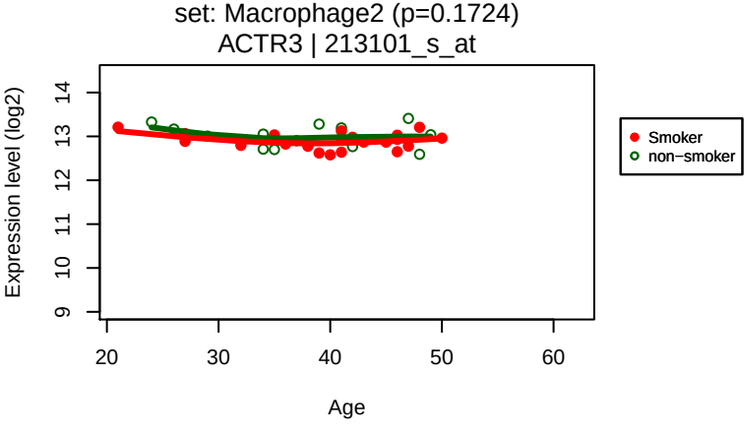

set: SmallAirway2 (p=0.0103)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker

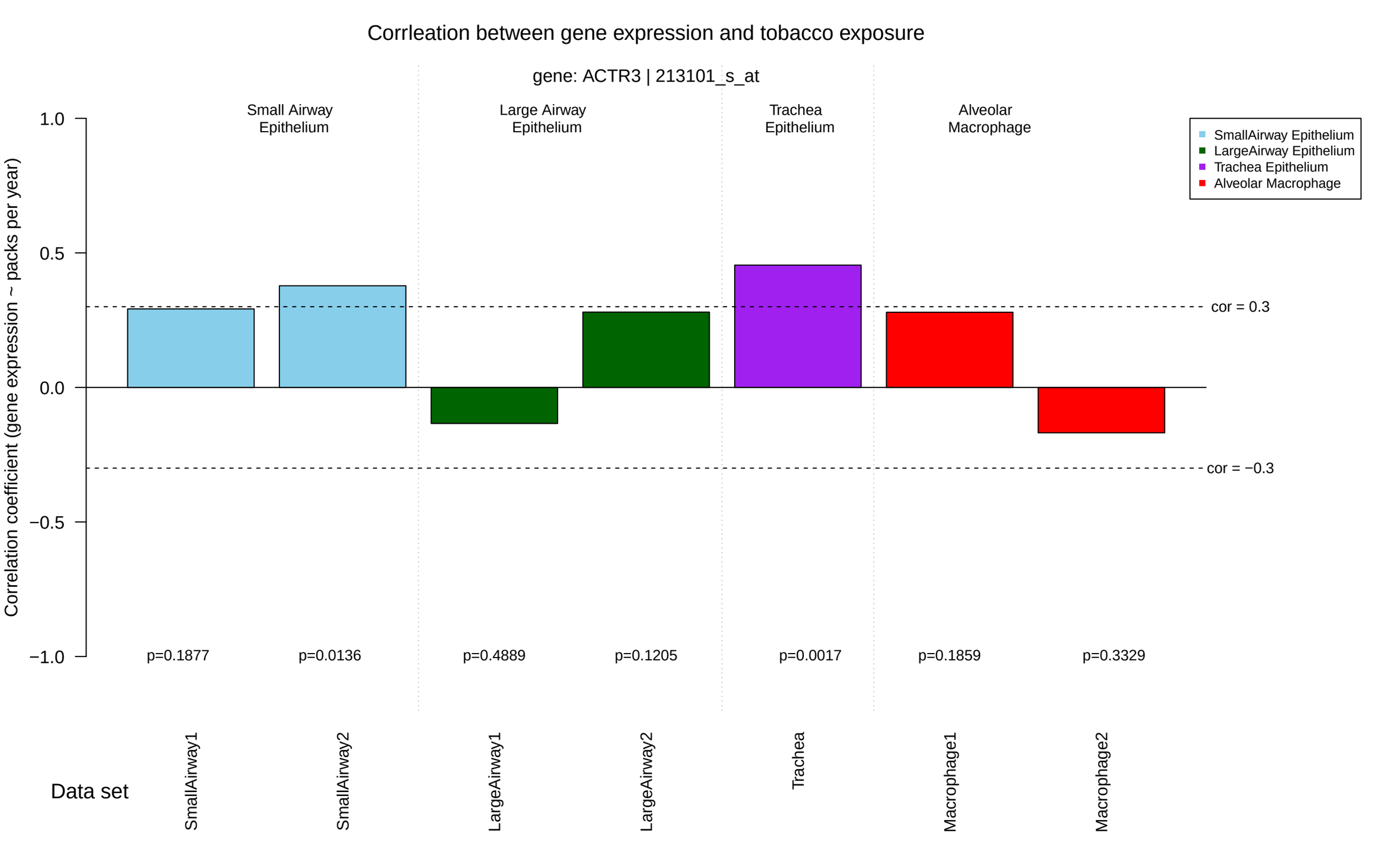

set: LargeAirway1 (p=0.1042)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0415)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0057)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4352)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1724)
ACTR3 | 213101_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR3 | 213101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1877
p=0.0136
p=0.4889
p=0.1205
p=0.0017
p=0.1859
p=0.3329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2